

Bitcoin Volatility Prediction

Forecasting 7-day realized volatility, not price direction

73,000+ Hourly BTC/USDT Records

How Bitcoin's Price Swings

Big rallies, sharp crashes, and everything in between

Bitcoin's price is highly non-stationary: since 2017 it has drawn down more than 80% twice, then rallied to new all-time highs both times. These swings - not day-to-day direction - are the volatility this project forecasts.



WHAT THIS MEANS

These price swings are volatility - the dispersion of returns, not their direction. This project forecasts the magnitude of that dispersion 7 days ahead.

What Is Volatility, and Why Predict It?

Volatility measures the dispersion of price movements over time, independent of direction. Low volatility means small, tight moves; high volatility means large, rapid swings either way.

AN ANALOGY

Predicting price is like guessing exactly where a boat will be tomorrow - nearly impossible. Predicting volatility is like guessing how rough the sea will be. That's a more tractable question, and usually the more useful one for risk management.

Why Predicting It Matters

Who Uses It	Why It Matters
Options traders	Required input for Black-Scholes σ
Risk managers	Scales Value-at-Risk estimates
Traders	Position sizing under changing risk
Exchanges	Sets margin/collateral requirements

HOW IT'S CALCULATED

$$\sigma_{7d} = \sqrt{\frac{1}{N-1} \sum_{i=t-N+1}^t (r_i - \bar{r})^2}$$

r = hourly return, $\bar{r}$ = its average, N = 168 hours (7 days)

INTUITION

The standard deviation of the last 168 hourly returns. A wider spread around the mean means a more turbulent week; a tighter spread means a calmer one.

Problem Statement

From a vague question to a testable target

THE AIM

Predict Bitcoin's 7-day realized volatility, 168 hours ahead of the features used to predict it - not the next hour's value. This gives traders and risk desks a continuously-updating volatility estimate, usable directly in position-sizing and margin decisions.

Target Definition

Target	volatility_7d, 168h ahead of the features
Horizon	168 hours (7 days) - never the next hour
Features	Lagged price/volume history
Success bar	Beat naive persistence & Holt-Winters

A NUMBER, NOT A CATEGORY

Downstream uses - option pricing, Value-at-Risk, margin - need a continuous estimate, not a high/low label. That rules out classification: this is a regression problem.

Data Overview & Feature Dictionary

DATASET: 73,000+ hourly BTC/USDT records

FEATURES: Price, volume, and engineered volatility measures

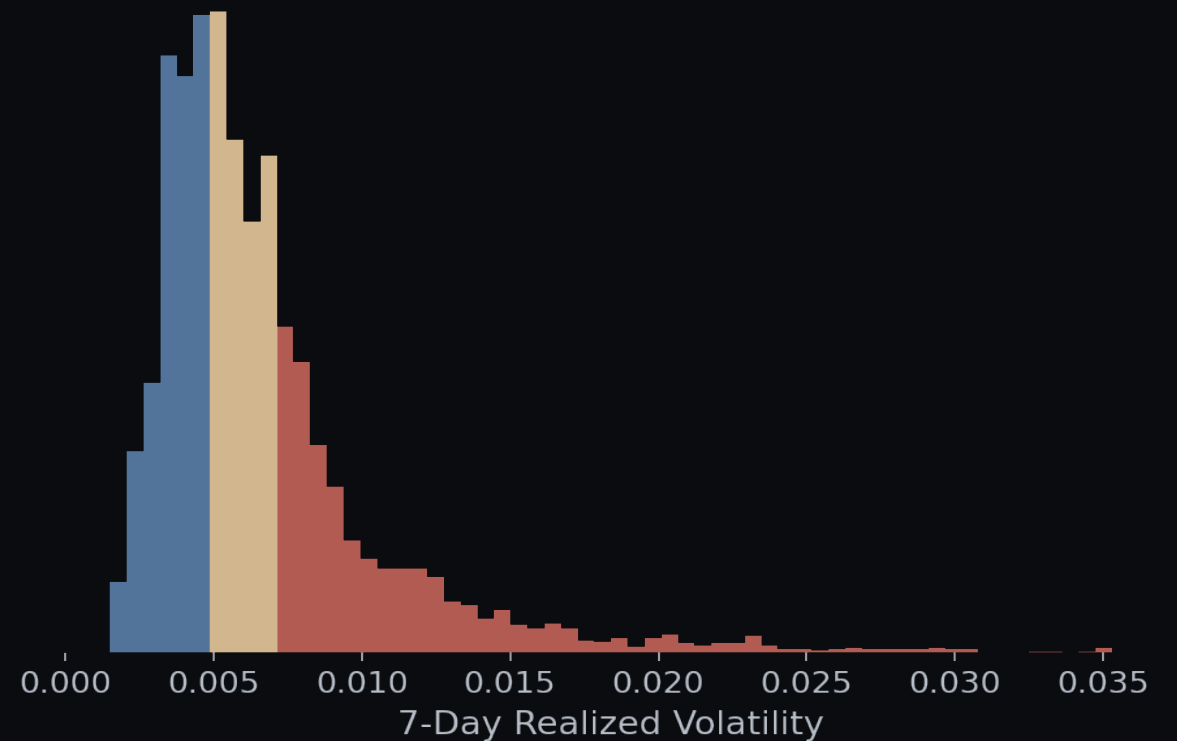
TARGET: Volatility over the next 7 days

Feature	Description	Type
RV_day / RV_week / RV_month	Realized volatility over the past day/week/month (HAR horizons)	Numeric
ATR_X	Average True Range, as a % of price	Numeric
EWMA_X	EWMA-smoothed volatility estimate	Numeric
SKEW_X	Realized skewness of hourly returns	Numeric
hourly_return	Hour-over-hour % price change	Numeric
volatility_7d (target)	7-day realized volatility, 168h ahead of the features	Target

Dataset & Data Cleaning

Data Cleaning Steps

- 1 Fixed 28 timestamp gaps in the trading data
- 2 Guarded zero-volume candles before rolling calculations
- 3 Standardized hourly timestamp frequency, no duplicates
- 4 Dropped ~30 days of rolling-window warm-up rows



Regime Mapping

Level	Regime	Volatility Range
0	Low	≤ 0.0047
1	Moderate	0.0047 - 0.0070
2	High	> 0.0070

Feature Engineering

Turning raw price data into usable signals

01

HAR Horizons

Aggregated realized volatility at day/week/month windows to capture the long-memory pattern the realized-volatility literature expects (Corsi, 2009).

02

ATR Scale Fix

ATR was on a ~45,000x different scale from every other regressor - a genuine bug. Fixing it was the single biggest improvement in the project.

03

EWMA Smoothing

Backtested the industry-standard exponential decay factor (λ) and added it as its own regressor, not just a separate model.

04

Realized Skewness

Captures return-distribution shape, not just magnitude - a genuinely different signal from the volatility-level regressors above.

EDA: Temporal Features

Is autocorrelation present - and how was it handled?

Diagnosing Autocorrelation

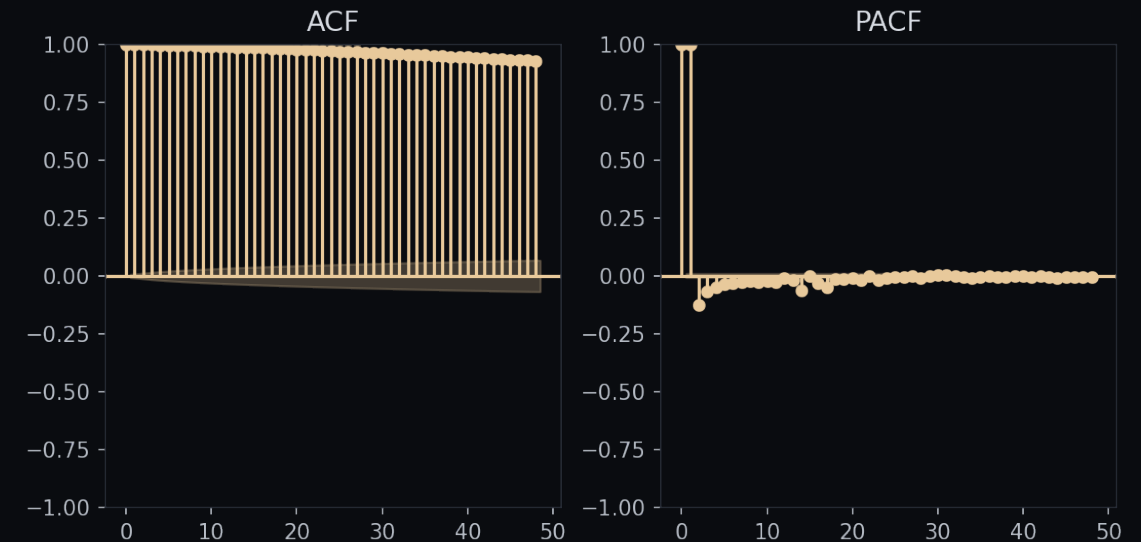
PRESENT - CONFIRMED

The ACF decays slowly across many lags (still ~ 0.9 at 48 hours) instead of dropping to zero, and the PACF shows that signal concentrated almost entirely at lag 1. This is genuine, strong autocorrelation - volatility clustering - not noise.

HOW IT WAS RESOLVED

Autocorrelated data breaks the i.i.d. assumption behind random splits and K-fold CV - a random shuffle would leak adjacent-in-time rows into training. Resolved with a chronological train/test split and TimeSeriesSplit CV (Preprocessing), plus lagged features - and by using the autocorrelation itself as HAR-RV-X's core signal, not discarding it.

ACF / PACF (48 Lags)



TAKEAWAY

Autocorrelation isn't a nuisance to remove here - it's the predictive signal. HAR-RV-X's lagged, multi-horizon regressors exploit it directly, while chronological splitting keeps evaluation honest.

Feature Selection: Correlation & Effect Sizes

Correlation Analysis: all continuous features tested against the target - the strongest are theoretically motivated, not just statistically convenient.

Retained (used by HAR-RV-X)

RV_day / RV_week / RV_month

Theoretically motivated (Corsi 2009)

ATR_X

$r = 0.142$, scale-fixed

EWMA_X

backtested smoothing

SKEW_X

distribution shape, not magnitude

Dropped (weak or redundant)

Other technical indicators tested

$r = 0.004$ - 0.013 , no meaningful signal

volume_chg

$r = 0.002$, no measurable relationship

Close / ma_7d (raw levels)

$r = -0.26$, non-stationary, spurious

CRITICAL FINDING

Even the best feature kept (ATR, $r = 0.142$) has only a modest connection to the target. This is why R^2 tops out around 0.24, not 0.9 - a ceiling set by the data itself, not by model choice.

Preprocessing

Preparing a leak-free feature matrix and target for modeling

Preprocessing Steps

- 1 Lag every feature 1-3 hours - only pre-prediction-time information is used
- 2 Build target: volatility_7d shifted 168h (HORIZON) into the future
- 3 Feature matrix uses lagged columns only - never same-time columns
- 4 Chronological 80/20 train/test split - never randomly shuffled
- 5 TimeSeriesSplit (5-fold) for every hyperparameter search - never random K-fold

Train/Test Split

TRAIN SET

58,352 rows

2017-08 to 2024-04

TEST SET

14,589 rows

2024-04 to 2025-12

WHY CHRONOLOGICAL

Time-series data is autocorrelated - a random shuffle would leak information from adjacent-in-time rows into training. Every split (train/test, and every cross-validation fold) respects chronological order.

NO FEATURE SCALING NEEDED

GARCH and EWMA fit directly on returns; HAR-RV-X's regressors are already on a comparable scale - no StandardScaler required anywhere in this pipeline.

Model 1: GARCH

GJR-GARCH with Student-t Errors (Tuned)

FORMULA

$$\sigma_t^2 = \omega + (\alpha + \gamma I_{t-1}) \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$$

Model Configuration

Vol Model	GJR-GARCH
Distribution	Student's t
p / o / q	1 / 1 / 1
AIC (Tuned)	96,564.5

Performance (Historical Test)

R2 (Historical)

-5.02

RMSE

0.003876

MAPE

90.54%

KEY INSIGHT

GARCH's forecast mean-reverts toward a long-run average well before 168 hours are up - a structural weakness at this horizon, not an implementation bug. It degrades further on live data (R2 = -110.68).

Model 2: HAR-RV-X

Ridge-Regularized Regression on RV Horizons (Final)

FORMULA

$$RV_{t+h} = \beta_0 + \beta_1 RV_{day} + \beta_2 RV_{week} + \beta_3 RV_{month} + \beta_4 ATR_X + \beta_5 EWMA_X + \beta_6 SKEW_X + \varepsilon$$

Model Configuration

Regressors	6 (RV horizons, ATR_X, EWMA_X, SKEW_X)
Regularization	Ridge
Alpha	0.01
Target Transform	log(target)

Performance (Historical Test)

R2 (Historical)

0.242

RMSE

0.001367

MAPE

22.83%

KEY INSIGHT

Best performer on both historical and live data (R2 = 0.178 live). A real scale bug fix in ATR_X (a ~45,000x mismatch) was the single largest driver of this result.

Model 3: EWMA

RiskMetrics-Style Exponential Smoothing (Backtested)

FORMULA

$$\sigma_t^2 = \lambda \sigma_{t-1}^2 + (1 - \lambda) r_{t-1}^2$$

Model Configuration

Named After	RiskMetrics (J.P. Morgan, 1996)
Lambda (Backtested)	0.995
Search Range	0.85 - 0.995
Selection	TimeSeriesSplit CV

Performance (Historical Test)

R2 (Historical)

0.059

RMSE

0.001532

MAPE

23.71%

KEY INSIGHT

A single, fully transparent parameter - no hidden layers, nothing opaque. A solid runner-up, and the risk desk's own trusted estimator, but not as sharp as HAR-RV-X's multi-horizon regressors.

Model Comparison

Historical Test-Set Performance Across All Models

Model	RMSE	R2	MAPE	Beats Persistence?
Naive Persistence (baseline)	0.001649	-0.090	24.41%	Baseline
Holt-Winters (baseline)	0.002845	-2.245	43.20%	No
GARCH (GJR, Tuned)	0.003876	-5.023	90.54%	No
HAR-RV-X (Final)	0.001367	0.242	22.83%	Yes
EWMA (lambda=0.995)	0.001532	0.059	23.71%	Yes

KEY FINDING

HAR-RV-X wins on every metric, beating naive persistence by 17.1% lower RMSE. Both HAR-RV-X and EWMA clear the persistence bar; GARCH and Holt-Winters do not. This gap holds up on genuinely unseen live data too, not just this historical split.

Limitations

Category 01

Data Constraints

- No fundamental or on-chain signals - only price/volume history.
- News, regulation, and macro shocks are invisible to the model.
- Single asset - no cross-crypto spillover signal used.

Category 02

Modeling Constraints

- GARCH's mean-reversion is structural, not tunable away.
- 22 robustness checks tested; most did not beat HAR-RV-X.
- Hardware/time limited exhaustive deep-learning exploration.

Category 03

Statistical Ceiling

- Feature correlations with the target are inherently weak.
- Models cannot exceed the information content of their inputs.
- A modest R2 ceiling is a predictable result, not a bug.

KEY TAKEAWAY

Meaningful improvement needs genuinely new information (news sentiment, on-chain metrics), not further tuning of what's already here. Neither more model complexity nor more compute alone is sufficient.

Operational Implications

Where a 0.24 R2 forecast is genuinely useful - and where it is not.

Appropriate Uses

Position Sizing

Scale trade size to the forecast - bigger expected move, smaller position.

Options Pricing Input

One data-driven volatility estimate to compare against market pricing.

Risk Reporting

A continuous VaR input for weekly risk reviews, not a single trade call.

Inappropriate Uses

Sole Trading Signal

Explains ~24% of variance - never trade on this number alone.

Real-Time Execution

Not built or tested for split-second, automated trade triggers.

Certainty in Legal/Compliance Filings

A forecast with real error bars, not a guarantee.

One input, not the final answer.

Thank You

Any Questions?